Optimized Statistical NBA Tool

Software Test Plans

DATE

April 27, 2026

TEAM

McGuire / Petrovic / Singh / Moua / Muterspaugh

DOCUMENT

Test Plans 001 – 003

Project Title: Optimized Statistical NBA Tool

Test Plan 001

Derived Metrics — Hit Rate Function · Unit Test, Logic

Test Objectives

Verify that the derived metrics function correctly computes the hit rate of a player's stat line over the last N games, returning accurate values for normal data, edge cases at the prop line boundary, and degenerate input (zero games played).

Test Approach

Unit test, logic verification. Three hand-traced test cases will be executed against the `hit_rate()` function in `analytics.py`:

- **Normal case** — player with 10 game logs, prop line set to a value cleanly above or below their average; verify the returned percentage matches the manual calculation.
- **Boundary case** — game stat exactly equal to the prop line; verify the function applies the strict over/under definition consistently (over = strictly greater than line).
- **Empty case** — player with zero game logs in the database; verify the function returns a safe value (e.g., zero or null) instead of raising a divide-by-zero or returning a misleading percentage.

Manual or Automated Test

Manual — three scripted logic traces executed against seeded test inputs.

Test Tools

Python interpreter (REPL), local SQLite database with seeded test rows, hand-calculated expected results.

Test Environment

Tester's local development machine running the Flask application against a controlled SQLite database.

Test Criteria

Requirement **1.2.2 (Derived Metrics)** — function must compute hit rate accurately across normal, boundary, and empty inputs.

Test Schedule

April 27, 2026

Test Team

- **Tester:** Jack Muterspaugh
- **Developer (excluded from testing):** Joey Petrovic — author of derived metrics logic

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Test Plan 002

Backend API ↔ Frontend Prop Screener · Integration Test, Interface

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Test Objectives

Verify that the backend API routes correctly fetch live NBA stats and odds from external services, parse the responses without data loss or malformation, and pass the resulting data accurately to the frontend prop screener for display.

Test Approach

Integration test, interface verification. The tester will:

- Issue requests to the backend API endpoints directly using Postman, inspect the JSON response structure, and confirm that fields, types, and values match the upstream API contracts.
- Trigger error conditions (invalid endpoint parameters, simulated upstream timeout) and verify the backend returns appropriate error responses rather than crashing.
- Load the prop screener page in the browser and use DevTools (Network tab) to confirm the frontend receives the same data the backend produced and renders it correctly.

Manual or Automated Test

Manual — performed through Postman and browser DevTools.

Test Tools

Postman (API request inspection), Chrome DevTools (Network and Console tabs), the running Flask development server.

Test Environment

Tester's local development machine running the full Flask application with valid Odds API credentials.

Test Criteria

- **1.1.1** — Backend route for data must be reachable and return valid responses
- **1.1.2** — API error handling must return a non-crashing error response on upstream failure
- **1.4.1** — Connection to The Odds API must succeed and return parseable JSON
- **1.4.2** — Odds data parsing must correctly extract player, market, line, and price fields without loss

Test Schedule

April 27, 2026

Test Team

- **Tester:** Gurjot Singh
- **Developers (excluded from testing):** Jack Muterspaugh (backend routes, nba_api), Jackson McGuire (Odds API integration, screener)

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Test Plan 003

Full User Workflow — Cross-Browser & Cross-Device · System Test, End-to-End Scenario

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Test Objectives

Verify the full end-to-end user workflow operates correctly across browsers and screen sizes: a user can search for a player, the prop screener loads with stats and live odds, and the prop detail view shows the player's recent game logs and visual trend charts without errors or unacceptable latency.

Test Approach

System test, end-to-end scenario. The tester will execute the following user flow on each target environment:

- Open the homepage and confirm today's games and the prop screener load within an acceptable response time.
- Use the search bar to look up a known active player and confirm the search resolves to that player's detail page.
- On the player detail page, confirm season averages, recent game logs, and Chart.js trend charts render correctly against current data.
- Navigate to a game detail page, confirm props for both teams render with correct hit-rate values.
- Repeat the entire flow on each target browser and viewport size; record any visual breakage, broken navigation, missing data, or response-time issues.

Manual or Automated Test

Manual — scenario-based exploratory testing performed by a non-frontend-developer team member.

Test Tools

Chrome (latest stable), Firefox (latest stable), Chrome DevTools device-emulation mode for mobile viewport testing, stopwatch or DevTools Performance tab for latency observation.

Test Environment

Production deployment of the Optimized Statistical NBA Tool, accessed over the public Replit deployment URL. Testing performed on:

- **Desktop** — Chrome and Firefox at standard 1920×1080 resolution
- **Mobile** — Chrome DevTools mobile emulation (iPhone and Pixel viewports)

Test Criteria

- **1.0.3** — User can navigate the application end-to-end without dead links or broken pages
- **1.0.4** — Application surfaces live data on every page that requires it
- **1.2** — Statistical data and derived metrics are present and accurate on user-facing pages
- **1.4** — Live odds data is present and accurate on user-facing pages
- **2.1 (Speed)** — Page load and interaction response times remain within acceptable bounds

- **2.2 (Cross-browser/device)** — Full workflow operates correctly across target browsers and screen sizes

Test Schedule

April 27, 2026

Test Team

- **Tester:** Ywjfeej Moua
- **Developers (excluded from testing):** Gurjot Singh (frontend styling and framework), Jackson McGuire (frontend pages and screener)